

MADIHA UMAR SHAIKH

Robotics & Automation Engineering Student | Computer Vision, ML & CAD Design
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SUMMARY

Motivated fourth-year Engineering student with a strong interest in Computer Vision, Machine Learning, and CAD design. Proficient in Python programming and experienced in creating 3D models using Fusion 360 and SolidWorks (basic). Passionate about developing innovative engineering solutions by combining AI, robotics, and design. A quick learner with strong problem-solving abilities, effective communication skills, and a collaborative mindset. Seeking opportunities to apply technical knowledge and gain hands-on industry experience.

SKILLS

Programming & Tools: Python (basic), MATLAB, Fusion 360, SolidWorks (basic), OpenCV, YOLO

Robotics & Automation: Robotics and automation problem solving using ladder logics in PLC, understanding of sensors and actuators, Arduino programming, embedded systems

Mechanical & Fluid Systems: Mechanical design and analysis, hydraulics, pneumatics, production and quality awareness, robot kinematics and joint behaviour

PROJECTS

Real-Time 3D Pose Estimation Using ArUco Markers

2025

- Developed a computer vision system using Python and OpenCV to detect ArUco markers and estimate their real-time X, Y, and Z coordinates relative to the camera.

Air Quality Monitoring System using Arduino

- Developed an air quality monitoring system to measure environmental parameters.
- Interfaced gas sensors with Arduino to monitor air quality levels and displayed sensor data for basic analysis and awareness.

Robotic Arm using Servo Motor (Academic Project)

2023

- Designed and built a robotic arm using servo motors for precise multi-axis movement and object manipulation.
- Programmed control logic using Arduino to achieve accurate positioning and smooth motion.
- Tested and optimized performance for stability, repeatability, and real-world pick-and-place tasks.

EDUCATION

B.E. in Robotics and Automation Engineering — Zeal College of Engineering & Research (S.P. Pune University)

Pursuing 4th Year | Aggregate CGPA to date: **8.67**

Sem I	Sem II	Sem III	Sem IV	Sem V	Sem VI
SGPA 6.32	SGPA 7.86	SGPA 8.23	SGPA 8.73	SGPA 8.86	SGPA 9.00

HSC: 51% (2023) SSC: 81% (2021)

EXTRA-CURRICULAR RELEVANT TRAININGS

- Exposure to cylinders, compressors, pumps, and control valves.
- Introduction to the 4/2 and 3/4 directional control valves (DCV).
- Designed and simulated robotic mechanisms using Robo Analyzer software.
- Mechanical designing.
- Analysed robot motion, kinematics, and joint behaviour.

EXPOSURE

Participated in technical events and robotics competitions at college and inter-college level, which helped develop team working skills and inter-personal communications.

CERTIFICATIONS

- OpenCV Bootcamp Certificate
- MATLAB Onramp
- MATLAB Simulink

MY INTEREST

I want to upgrade my knowledge in Computer Vision, Machine Learning, and want to know more about Mechanical (CAD) Designing. I will be grateful if I come across new domains and will work on it effortlessly during my industrial practical training, which I am seeking.